

in exposed occupations. While their analysis covers 25 broad occupation groups, we exploit variation across a much wider variety of job types, as well as a firm-level analysis that examines within-firm dynamics that would be invisible in occupational aggregates. In addition, our individual-level posting data enables examination of a wider set of outcome variables such as salary distributions and seniority-specific employment, as well as heterogeneity in treatment effects by pre-treatment compensation or location.

Taken together, our findings raise important policy questions. The disproportionate elimination of entry-level positions disrupts the conventional model of skill development, where employees master progressively complex tasks through hands-on experience with routine work. Without junior roles that serve as training grounds, firms may struggle to develop future senior talent internally, creating a hollowed-out labor market where entry becomes increasingly difficult (Ide, 2025). This dynamic could exacerbate inequality as those with pre-existing experience benefit from reduced competition while new entrants face shrinking opportunities. Policy responses must address both immediate displacement and longer-term skill development pathways.

2 Data

Our primary data comes from Revelio Labs, which aggregates and processes publicly available employment information from professional networking platforms, company websites, and job boards using machine learning algorithms to create proprietary longitudinal labor market databases. Revelio Labs transforms semi-structured public profile information into a structured panel dataset through systematic parsing, standardization, and cleaning procedures. Their data have been validated against official statistics and used in recent academic research on labor market dynamics (Fadhel et al., 2021; Eisfeldt et al., 2023; Hampole et al., 2025). In Appendix A4, we separately validate that Revelio Labs employment and job posting data closely track Office for National Statistics aggregates both over time and across industries. We use two separate Revelio Labs datasets: work profiles and job postings.²

²The data used in this paper were downloaded on 12 August 2025.

2.1 Job Profile Data

The first dataset consists of all publicly available professional LinkedIn profiles between January 2021 through June 2025. The dataset includes approximately 75 million employment spells from 23 million unique individuals across 2.5 million UK-based firms. Each observation includes the worker’s job title, employer, start and end dates (if available), seniority level,³ and compensation.⁴ The data provide near-real-time employment stocks, which enable us to track within-firm workforce composition changes over time.

From the panel of individual LinkedIn profiles, we construct a monthly firm-level panel from January 2021 through June 2025. For each firm-month, we calculate total employment as the count of unique individuals with active employment spells at a certain company in that month, and log-transform this variable. This is our main employment stock measure. We then separate total employment into junior and senior employment stocks based on Revelio seniority classifications of jobs. We also calculate average compensation as the log of the mean of reported salaries plus bonuses in a firm. We only focus on firms that remained in business throughout the sample period.

To calculate firm-level exposure scores, we use the assessment framework developed by Eloundou et al. (2024). They evaluate 19,265 detailed work activities and tasks from the O*NET 27.2 database to determine whether LLM assistance could reduce task completion time by at least 50% while maintaining quality. We focus on their ‘E1’ measure, which captures exposure to current LLMs, rather than LLMs combined with additional software. We do so because it is the most appropriate metric for identifying short-term labor market effects that do not require additional software development or complex organizational changes. To obtain our firm-level

³Revelio Labs determines seniority using an ensemble model that combines three components: (i) current job characteristics including title, company, and industry; (ii) employment history including tenure and previous position seniority; and (iii) individual age. These three scores are averaged to create a continuous seniority metric, which is then mapped to seven ordinal levels based on keyword benchmarks (e.g., “junior,” “senior,” “director”). The levels range from Entry Level (e.g., interns, trainees) through Junior, Associate, Manager, and Director levels, to Executive and Senior Executive levels (e.g., C-suite positions).

⁴Revelio Labs estimates compensation using a predictive model trained on publicly available visa application data, self-reported salaries, and job posting information. The model incorporates position-specific variables (job title, seniority level, company, location), individual-specific factors (tenure at company), and temporal controls (year of observation) to generate salary predictions for positions where compensation data are not directly observed.

measures, we first obtain occupation-level measures by calculating the share of tasks within an occupation that meet the 50% time-saving threshold. We weight this by task importance, assigning 0.5 to supplementary tasks and 1.0 to core tasks.⁵

Formally, for occupation o :

$$\alpha_o = \sum_{t \in T_o} w_t \cdot \mathbb{1}_{[\text{LLM reduces time by } \geq 50\%]} \quad (1)$$

where T_o denotes occupation o 's task set, w_t represents task importance weights (1.0 for core tasks, 0.5 for supplemental), and $\mathbb{1}_{[\cdot]}$ is an indicator function for substantial LLM impact. α_o gives us an occupation-level exposure score that we can assign to each employee.

We then calculate each firm's average exposure score by taking the average exposure score of all employees who report working for that firm on 30 October 2022, immediately before the introduction of ChatGPT.⁶

Formally:

$$\text{Exposure}_i = \sum_o \frac{L_{io, \text{Oct2022}}}{L_{i, \text{Oct2022}}} \cdot \alpha_o \quad (2)$$

where $L_{io, \text{Oct2022}}$ denotes firm i 's employment in occupation o as of October 2022.

We restrict our sample of firms to those with at least 100 employees immediately before ChatGPT's introduction to address two methodological concerns. First, larger firms reduce noise from idiosyncratic employment changes and ensure adequate LinkedIn coverage for constructing reliable firm-level exposure scores from individual employee profiles. Second, this threshold minimizes attrition bias from firms exiting the sample while maintaining a sufficiently large set of employers that account for a large part of UK employment. Appendix A3.3 in the Appendix examines the sensitivity of our results to different firm-size thresholds.

Our final firm-level sample includes 26,979 firms with 1,268,013 firm-month ob-

⁵Our robustness checks cover alternative weights based on O*NET task-importance classifications.

⁶We measure firm-level exposure using the pre-treatment workforce composition to avoid endogeneity concerns. If we calculated exposure using post-ChatGPT employment, our measure would be contaminated by the treatment effect itself, as firms might have already adjusted their occupational mix in response to LLM capabilities. By fixing exposure at the pre-treatment date, we ensure that our exposure measure captures firms' initial vulnerability to LLM disruption rather than their endogenous response to it.

Table 1: Summary Statistics Firm Employment

	Low exposure		High exposure	
	Before	After	Before	After
Total jobs	456	507	639	708
Low seniority (%)	0.705	0.703	0.667	0.662
High seniority (%)	0.295	0.297	0.333	0.338
Compensation	£ 94,325.66	£ 94,079.89	£100,230.08	£100,184.41
LLM exposure score	0.096	0.096	0.172	0.172

Notes: This table reports summary statistics on firm-level employment outcomes by exposure to large language models (LLMs). Firms are classified into high- and low-exposure groups based on whether their average exposure score lies above or below the median across all firms. *Total jobs* gives the mean number of active positions in the firm. *Low seniority* and *High seniority* denote the share of employees in junior and senior roles, respectively, as defined by seniority of 1-3 (junior) or 4-7 (senior). *Compensation* gives the mean annual salary plus bonuses, and *LLM exposure score* is the average predicted LLM-exposure of jobs in a firm based on Eloundou et al. (2024). *Before* refers to the pre-treatment period (prior to November 2022), while *After* refers to the post-treatment period (November 2022 onward).

servations. These firms employ 16 million workers in our data as of October 2022, representing approximately 49% of UK employment (Office for National Statistics, 2022).

Table 1 presents firm-level employment summary statistics. Firm-level employment stocks increased for both exposure groups following ChatGPT’s introduction, rising 11.2 percent for low-exposure firms (from 456 to 507 employees) and 10.8 percent for high-exposure firms (from 639 to 708 employees). However, this increase could reflect an increase in the number of people joining LinkedIn during the sample period. The workforce composition reveals important differences: high-exposure firms maintain a more senior-skewed workforce, with 33.3 percent of positions at high-seniority levels compared to 29.5 percent in low-exposure firms. Following ChatGPT’s release, this seniority gap widened slightly, as the share of high-seniority positions increased by 0.5 percentage points in high-exposure firms versus 0.2 percentage points in low-exposure firms. These patterns suggest that high-exposure firms are gradually shifting their workforce composition toward more senior roles, potentially reflecting the displacement of entry-level positions. Consistent with their more senior workforce composition, high-exposure firms offer higher average compensation.